

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	30 September 2026
Team ID	SWTID-2026-4452
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Technical Architecture

The architecture represents the end-to-end process described in the project source: spreadsheet preparation, ServiceNow data loading, Import Set staging, Transform Map field mapping, transformation, validation, duplicate prevention through Coalesce, reporting, and dashboard visualization.

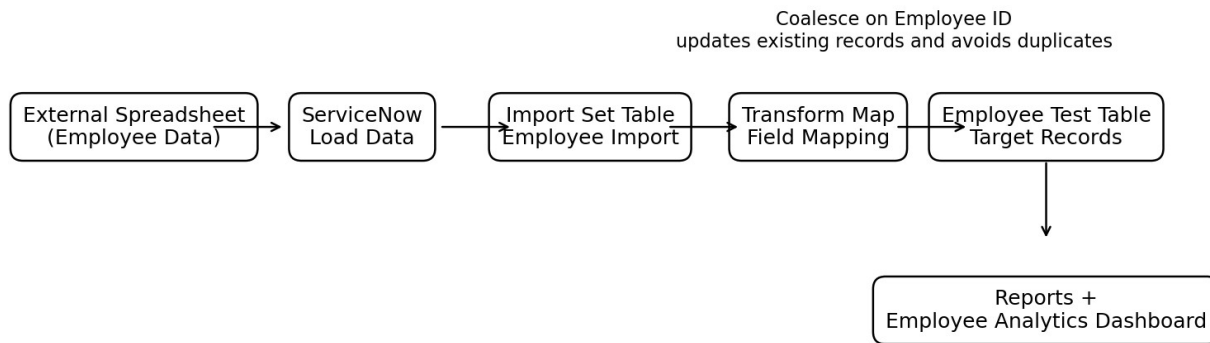


Table-1 : Components & Technologies

S.No	Component	Description	Technology
1	User Interface	User interacts with the ServiceNow platform to load spreadsheet data, configure mappings, run transforms, and view reports/dashboard.	ServiceNow user interface
2	Application Logic-1	Loads spreadsheet data into the staging area before transformation.	ServiceNow Load Data / Import Set
3	Application Logic-2	Maps source spreadsheet fields to corresponding target table fields.	ServiceNow Transform Map + Mapping Assist / Auto Map Matching Fields
4	Application Logic-3	Prevents duplicate records and updates existing records when the configured employee identifier matches.	Transform Map Field Map with Coalesce
5	Database	Stores staged import data and transformed employee records.	Import Set table; Employee Test table
6	Cloud Database	Cloud database service is not described in the source project.	Not specified
7	File Storage	External spreadsheet is prepared/downloaded as an Excel file and selected during Load Data.	Excel (.xlsx) file / file upload
8	External API-1	No external API is described in the project workflow.	Not specified
9	External API-2	No second external API is	Not specified

S.No	Component	Description	Technology
		described in the project workflow.	
10	Machine Learning Model	No machine learning model is described in the project.	Not specified
11	Infrastructure (Server / Cloud)	The application workflow is implemented in ServiceNow; specific local/cloud server configuration is not stated.	ServiceNow platform; configuration not specified

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No open-source framework is specified in the project document.	Not specified
2	Security Implementations	The source does not specify encryption, IAM, firewall, or OWASP controls.	Not specified
3	Scalable Architecture	The project supports bulk spreadsheet import, transformation, duplicate prevention, reporting, and dashboard visibility. A specific scalability architecture is not stated.	Not specified
4	Availability	Load balancing or distributed-server configuration is not described.	Not specified
5	Performance	The project emphasizes efficient and accurate bulk data migration and transformation; specific requests/sec or caching configuration is not stated.	Not specified

Project Outcome / Key Capabilities

- Structured employee data can be imported from an Excel spreadsheet into ServiceNow using Import Sets and Transform Maps.
- Source fields are mapped to Employee Test fields: Employee ID, Employee Name, Email, Department, and Location.
- Coalesce on Employee ID prevents duplicate records and updates existing records.
- Reports provide department-wise distribution, location-wise distribution, and an employee list.
- Reports are consolidated into the Employee Analytics Dashboard for centralized visibility.

Source: Import Data using Transform Maps (Spreadsheet).